

Principles of Economics

Class 4: Owners, Managers and Employees

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The relationship between workers and the firm

Incomplete contracts

The labour discipline model: employment rent

The labour discipline model: the worker's best response

The labour discipline model: the firm's profit maximisation

The labour discipline model and unemployment

The principal-agent model

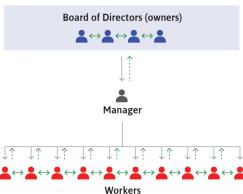
1. The relationship between workers and the firm

- ▶ Last class: conflicts of interest in social interaction.
 - ▶ Very relevant inside the firm.
 - ▶ **Owner and manager**: same objectives?
 - ▶ And **workers and managers**?
- ▶ Today: **understanding** (and resolving) **conflicts of interest within the firm**:
 - ▶ How are **wages** set inside the firm? And the effect on unemployment?
 - ▶ What do **firms' interactions** do to the economy as a whole?

Owners, Managers and Employees

An example of a conflict of interest within the firm

- ▶ Managers and workers are an input with objectives of their own.
- ▶ Inside the firm **several agents interact**:



- ▶ **Owners** seek to maximise profits;
 - ▶ **Managers** seek promotion, power and pay;
 - ▶ **Workers** seek to earn their wage with minimum effort.
-
- ▶ **Asymmetric information** between hierarchical levels (dotted upward arrows).

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- ▶ The firm's **profits belong to the owners**.
- ▶ **But managers generate it**, and do not automatically share in it.
- ▶ Two ways out of this **conflict of interest**:
 - ▶ Tie managers' pay to the share price;
 - ▶ Monitor the managers' performance.

By Chris Isidore

Acumula los gig
que te sobren
hasta 3 meses

Lancien in progress

MÓVEL 8GB

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simyo

2. Incomplete contracts

- ▶ **Hiring employees** is unlike buying any other input.
 - ▶ Labour cannot be bought, only rented.
 - ▶ The worker owns their actions, hard to pin down by contract (incomplete contracts):
 - ▶ **Uncertainty**: not every contingency can be foreseen, nor everything the worker will have to do.
 - ▶ **Measurement**: observing the worker's effort is costly, or impossible.
- ▶ Performance may be measurable but impossible to prove in court: **lack of verifiability**.
- ▶ Even with perfect foresight and observable effort, a **perfectly enforceable** contract covering everything would be **impossible to write**.

Is Quiet Quitting Real?

BY JIM HARTER



STORY HIGHLIGHTS

- At least half of the U.S. workforce is quiet quitting
- The workplace, amid the pandemic, got worse for younger workers
- Managers are essential to combatting quiet quitting

- ▶ "Quiet quitters" make up at least 50 percent of the U.S. workforce, probably more, according to Gallup. The trend toward quiet quitting (the idea spreading virally on social media that millions of **people are not going above and beyond at work and are strictly doing the minimum required** of their job) could get worse. This is a problem because most **jobs today require some level of extra effort** to collaborate with co-workers and meet customers' needs. Read the full article: [Gallup](#)
- ▶ Even in a job you like, effort brings **disutility**.

Incentivising effort: piece-rate contracts

Incentivising effort: pay for performance

- ▶ Effort is hard to observe, but **output can be measured**.
 - ▶ The everyday example: sales commissions.
- ▶ If **output tracks effort**:
 - ▶ Pay for performance: **a wage per unit of output**.
- ▶ Possible **problems**:
 - ▶ How can we **measure output**? Is it accurate?
Think about how a teacher assesses a student.
 - ▶ How do we measure **workers' contribution to output**?
Think about a team project you may have done. Did everyone contribute in the same way?
 - ▶ What about the **quality of output**?
Think about a 20-page essay – how do we check its quality without reading it?

► **Other incentives firms can offer:**

- **Fringe benefits:** health insurance, childcare, gym;
- **Promotion** on performance;
- Offer **adequate working conditions**;
- **Work–life balance** and flexibility: remote work, flexible hours;
- **Additional responsibilities.**

► And **workers** have **intrinsic incentives**:

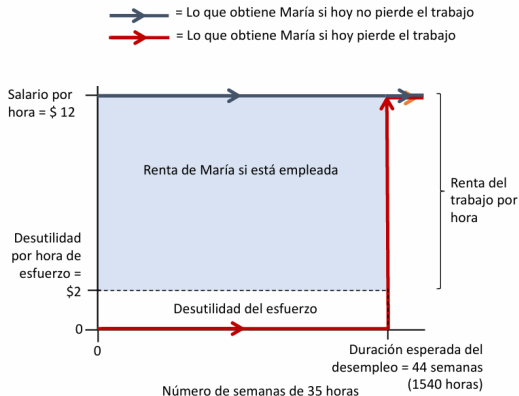
- Work ethic;
- Feelings of responsibility;
- Reciprocity for good working conditions;
- **Fear of being fired:**
Unemployment as a worker-discipline device (Stiglitz) / **reserve army of the unemployed** (Marx)

3. The labour discipline model: employment rent

- ▶ **Key assumption:** employees fear being fired when **they are paid more than they would get elsewhere:**
Employment rent = cost of job loss
- ▶ It is the opportunity cost (and the subjective cost) of not working.
- ▶ The **cost of job loss** includes:
 - ▶ Lost income while searching for a job;
 - ▶ Costs of starting again: relocation, new skills;
 - ▶ Loss of non-wage benefits;
 - ▶ Social costs (stigma of being unemployed).
- ▶ From the employment rent comes the **reservation wage:**
 - ▶ The wage that leaves the worker indifferent between working or not (**the next best option:** another job or benefits).
 - ▶ Effect on **effort and on the fear of being fired?**

► María's **employment rent** given a certain level of effort:

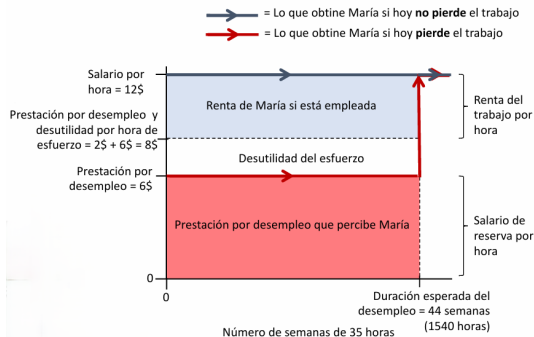
- With an hourly wage of \$12;
- With an **expected unemployment** duration of 44 weeks;
- In an economy **WITHOUT unemployment benefits**.



- **Employment rent/hour** = wage – disutility of effort = $\$12 - \$2 = \$10$
- **Total employment rent** = $\$10 \times 1540 = \15400

- ▶ María's **employment rent** given a certain level of effort:

- ▶ With an hourly wage of \$12;
- ▶ With an **expected** unemployment duration of 44 weeks;
- ▶ In an economy **WITH** unemployment benefits.



- ▶ **Employment rent** = wage – reservation wage – disutility of effort = $\$12 - \$6 - \$2 = \$4 < \$10$
- ▶ **Total employment rent** = $\$4 \times 1540 = \6160

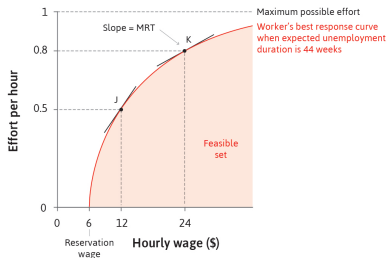
4. The labour discipline model: the worker's best response

- ▶ A higher cost of job loss means a **higher employment rent**, and **more incentive to exert effort**.
- ▶ So the **firm** can buy **more effort** by **raising the wage** (or **lowering the disutility of work**: better conditions).
- ▶ But both are **more costly** for the firm.
- ▶ **Game:**
 - ▶ **Players:** firms and workers
 - ▶ **Strategies:**
 - ▶ The employer sets a *wage* and fires only on clear evidence of low effort;
 - ▶ The worker chooses *effort*, knowing that less effort means more risk of dismissal.
 - ▶ **Payoffs:**
 - ▶ Employers: output of labour – wages
 - ▶ Workers: employment rents
- ▶ Here **workers** are supply (they sell labour) and **firms** are **demand** (they buy it).

The labour discipline model

The worker's problem

The worker picks **the effort that suits them at each wage, given the risk of dismissal**. In a graph:



- ▶ **Best response curve:** the most effort the worker will supply at each wage.
- ▶ **Feasible wages:** above the reservation wage.
 - ▶ Below it, the worker prefers not to work.

- ▶ The best response defines the **firm's feasible set**.
- ▶ **Points off the frontier are inefficient:**
 - ▶ The firm overpays for that effort;
- ▶ Points on the **frontier** are **efficient** / options with no waste;
- ▶ **The slope of the best response is the MRT**, positive and decreasing:
 - ▶ How much effort one more euro of wage buys.

5. The labour discipline model: the firm's profit maximisation

- ▶ To **maximise profits**, minimise production costs:

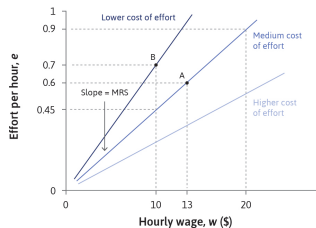
$$\begin{aligned}\text{Profit} &= \text{Revenue} - \text{Costs} \\ &= (\text{output} \times \text{price}) - [(\text{hours of work} \times \text{hourly wage}) + \text{other costs}] \\ &= [(\text{effort} \times \text{hours of work}) \times \text{price}] - [(\text{hours of work} \times \text{hourly wage}) + \text{other costs}]\end{aligned}$$

- ▶ The formula ties **wages to effort**:
 - ▶ Wages imply a cost;
 - ▶ But they also buy effort, and effort is output and revenue.
- ▶ So:
 - ▶ The firm picks the **feasible effort-wage combination with the lowest cost** per unit of effort.
 - ▶ That is, maximise e/w subject to the feasible set.

The labour discipline model

The firm's view

- ▶ **Isocost curves:** all effort-wage combinations that cost the same.
 - ▶ **The more vertical the line, the lower the cost.**
 - ▶ [more effort is obtained with a lower wage]
- ▶ The slope of an isocost is the **MRS**:
 - ▶ How much extra effort compensates the employer for one more euro of wage.

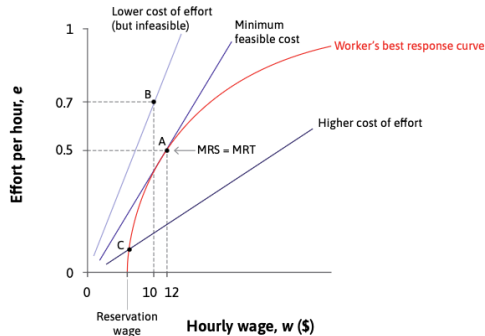


For the same level of effort, the more vertical lines require a lower wage.

The labour discipline model

Equilibrium

- ▶ **Profits are maximised** when:
 - ▶ the **isocost line** [lowest cost] is **tangent to the worker's best response** [most effort per euro of wage]
- ▶ This implies that, in **equilibrium**:
 - ▶ **MRS=MRT**
[tangency of the isocost / best response]
- ▶ In our example the equilibrium wage is \$12/hour:
 - ▶ The lowest-cost option that guarantees maximum effort.



6. The labour discipline model and unemployment

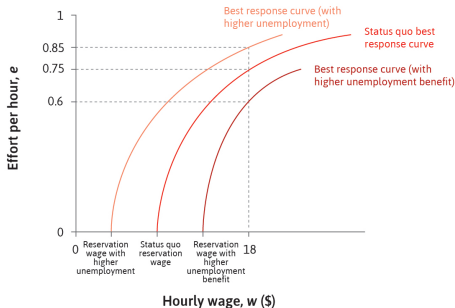
Involuntary unemployment

- ▶ The model assumes jobs are not recovered at once: there is a **spell of involuntary unemployment**.
 - ▶ **Involuntary unemployment**: without work, but wanting a job at the wage and conditions identical employed workers have.
- ▶ Suppose María finds an identical job the day after being fired [**no unemployment**].
 - ▶ Then **her employment rent is zero**: keeping or losing the job is the same to her.
 - ▶ Her best response is **zero effort**, and that **cannot be an equilibrium**:
 - ▶ The employer will not pay a wage to a worker who does nothing!
- ▶ In equilibrium, **wages and involuntary unemployment are both high enough** to make the **employment rent** worth protecting.

The labour discipline model

Changes in equilibrium

- ▶ The equilibrium hangs on the **wage-effort relationship**: the worker's best response.
- ▶ **Best response and equilibrium move** with changes in:



- ▶ The utility of the things the wage can buy;
- ▶ The disutility of effort;
- ▶ The reservation wage;
- ▶ The probability of being fired at each level of effort.

7. The principal-agent model

- ▶ Not only in employment. **Incomplete contracts** arise whenever:
 - ▶ The relationship spans a period of time;
 - ▶ There is **uncertainty**;
 - ▶ There are difficulties in **measurement**;
 - ▶ Information is not **verifiable**;
 - ▶ The **enforcement of laws** is unavailable or very difficult;
 - ▶ There is a **conflict of interest**.
- ▶ **Principal-agent models** describe the parties under incomplete contracts.
Examples:
 - ▶ Employment contracts: manager (principal) / workers (agent)
 - ▶ Investments: investors (principal) / fund managers (agent)
 - ▶ Elections: citizens (principal) / politicians (agent)
- ▶ **Agency relationship: the principal appoints the agent** to act on their behalf, possibly with **conflicting interests**.
- ▶ **With incomplete contracts** the agent's action is not fully observable:
hidden action.

Section by section (Micro, Unit 6)

1. **The relationship between workers and the firm** → owners, managers and workers pursue different objectives, with asymmetric information between levels.
2. **Incomplete contracts** → uncertainty, measurement and verifiability make effort impossible to write into a contract.
3. **Employment rent** → the cost of job loss: wage minus reservation wage minus the disutility of effort.
4. **The worker's best response** → the most effort supplied at each wage; its slope is the MRT and it bounds the firm's feasible set.
5. **Profit maximisation** → isocost lines are the firm's indifference curves; the equilibrium is tangency, $MRS=MRT$.
6. **The model and unemployment** → without involuntary unemployment the rent is zero and no effort is supplied: unemployment is part of the equilibrium.
7. **The principal-agent model** → the same hidden-action problem in investment, politics and beyond.

Key concepts: incomplete contract, verifiability, pay for performance, employment rent, reservation wage, disutility of effort, best response curve, isocost line, $MRS=MRT$, involuntary unemployment, principal-agent, hidden action.